

Frequency-Guided Semantic Steganography for High-Fidelity Self-Contained Style-Transfer Image Hiding

Ngoc-Giau Pham^{1,2}, Minh-Tuan Bui³, and Phuoc-Hung Vo^{1(*)}

¹ College of Engineering and Technology, Tra Vinh University, Vinh Long, Vietnam
hungvo@tvu.edu.vn

² Faculty of Engineering and Technology, Tien Giang University, Dong Thap, Vietnam
phamngocgiau@tgu.edu.vn

³ RMIT University, Ho Chi Minh City, Vietnam
bmtuan0704@gmail.com

(*) Corresponding author

Abstract. Style-transfer steganography uses the texture of stylised images to mask embedding residuals, but repeated stylisation can degrade recovery. We investigate two design choices that contribute to this trade-off: distortion regularisers without a target-fidelity threshold and spatially dense payloads. FGSS encodes the secret as a $\mathbf{z} \in \mathbb{R}^{64 \times 32 \times 32}$ semantic token and an auxiliary thumbnail. It applies a target-calibrated penalty above an MSE threshold derived from a desired cover PSNR, together with a wavelet-inspired term that discourages low-frequency residual energy. An oracle-guided reverse decoder and probabilistic future-pass supervision train the model for recovery after as many as three cumulative stylisation passes. On 600 COCO val2017 content–style pairs, FGSS obtains 39.61 ± 0.35 dB cover PSNR, 0.9914 SSIM, and 0.0195 LPIPS. Clean-channel reverse SSIM rises from 0.5234 for the content-free baseline to 0.8070 with the decoded token; an oracle-token diagnostic reaches 0.8136. In a reduced 192×192 single-pair ablation, the target-calibrated penalty increases cover PSNR by 5.76 dB. Token MSE differs by at most 1.2×10^{-4} across the integrated clean, Gaussian, and JPEG-style channels. The embedding path has 1.9 M parameters and runs in 0.0096 s per image on an RTX 5070 Ti. [Pretrained checkpoints, code, and split manifests are publicly available for full reproducibility.](#)⁴

Keywords: Image steganography · Neural style transfer · Frequency-domain learning · Capacity ceiling · Robustness · Semantic tokens.

1 Introduction

Deep image-hiding models can carry larger payloads than traditional least-significant-bit methods, but their residuals may remain detectable by learned

⁴ Code and checkpoints repository: <https://github.com/phamngocgiau/FGSS-ISDS2026>

steganalysers [1,2,3]. Style-transfer steganography instead uses the naturally variable texture of a stylised carrier [4,5,6,7,8,9,10,11]. In the self-contained setting [4], the receiver reconstructs the original content from the stego image and shared model weights. Published methods still show a marked trade-off between cover fidelity and recovery, particularly after a second or third stylisation pass [4,5].

We study two choices in this trade-off: a coefficient-weighted distortion loss with no target-fidelity threshold, and a payload that spans the carrier spatially. FGSS replaces the spatial secret with a fixed-size semantic token and thumbnail. Its capacity-ceiling loss is a soft, one-sided penalty above an MSE threshold computed from a target PSNR; a separate low/high-frequency proxy discourages residual energy in the low-frequency band. Oracle-guided reverse decoding and serial-consistency supervision are used to improve recovery after repeated transformations.

The full model has 10.7 M parameters, of which 1.9 M are used on the embedding path. On 600 COCO val2017 [12] pairs, it obtains 39.61 dB cover PSNR, 0.9914 SSIM [13], and 0.0195 LPIPS [14]. The contributions are: (i) a target-calibrated soft penalty on cover MSE; (ii) a differentiable frequency proxy that penalises low-frequency residual energy; (iii) a compact semantic-token payload; and (iv) reverse and serial-recovery experiments, including robustness and steganalysis probes. The reduced single-pair ablation changes cover PSNR from 35.61 to 41.37 dB when the target-calibrated penalty is enabled.

2 Related Work

2.1 Deep Image-in-Image Steganography

Recent image-hiding systems use invertible networks [1], normalising flows [2], attention modules [3], and frequency-domain objectives [15]. These models can carry full images or message grids, although learned detectors such as SRNet [16] and XuNet [17] may still identify residual statistics. FGSS does not claim undetectability; it uses a target-calibrated distortion penalty and reports detector results in Section 4.6.

2.2 Coverless and Generative Steganography

Coverless steganography and steganography without embedding generate or select containers through disentanglement models [18,19] or diffusion models [20]. Because these approaches do not add a conventional embedding residual, their payload definition and carrier control differ from residual-based image hiding. FGSS adopts a compact latent payload but retains a deterministic style-transfer carrier and a bounded-scale residual.

2.3 Style Transfer Steganography

Style-transfer steganography hides information in, or alongside, features used for artistic stylisation [4,21,5,6,7,10,11]. Existing systems use different carriers,

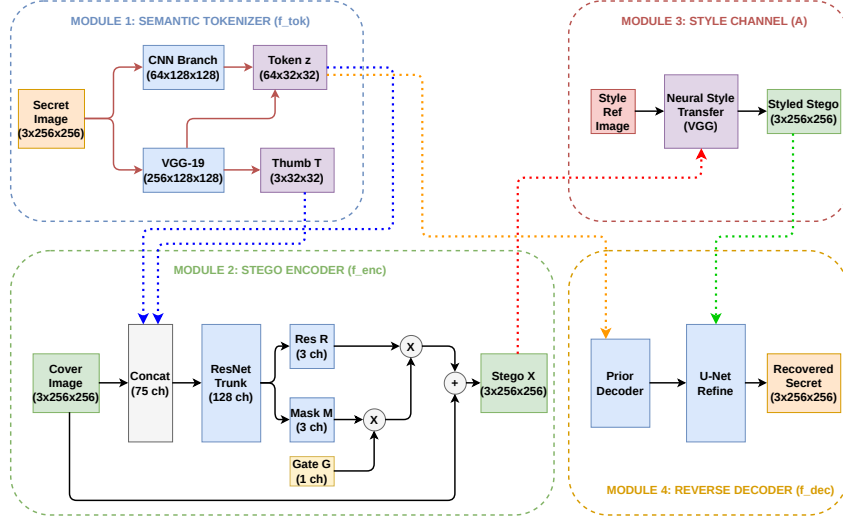


Fig. 1. Master architectural overview of the proposed Frequency-Guided Semantic Steganography (FGSS) framework. The diagram illustrates the data flow across four distinct modules enclosed in dashed bounding frames: (i) Module 1: Semantic Tokenizer f_{tok} ; (ii) Module 2: Stego Encoder f_{enc} with capacity ceiling & texture gate $\mathbf{G}$; (iii) Module 3: Style Transfer & Attack Channel $\mathcal{A}$; and (iv) Module 4: Reverse Decoder f_{rev} with oracle distillation. Feature map dimensions ($C \times H \times W$) are denoted within each functional block.

payloads, and fidelity targets, which makes direct PSNR comparison difficult. FGSS adds an explicit target-derived MSE threshold to the training objective; the threshold is a soft penalty rather than a hard guarantee on test images.

3 Proposed Method

3.1 Notation and System Overview

Scalars are plain math italic, vectors bold lower case, tensors bold upper case. The cover image is $\mathbf{X}_{\text{cover}} \in \mathbb{R}^{3 \times H \times W}$, the secret content is $\mathbf{C} \in \mathbb{R}^{3 \times H \times W}$ and the painting reference is $\mathbf{S}$. The cover is obtained by an optimisation-based stylisation operator $f_{\text{style}}(\mathbf{C}, \mathbf{S})$ held fixed during stego training. We write $\mathbf{z} \in \mathbb{R}^{d_{\text{tok}} \times g_{\text{tok}} \times g_{\text{tok}}}$ for the semantic token used as payload and $\mathbf{T} \in \mathbb{R}^{3 \times g_{\text{tok}} \times g_{\text{tok}}}$ for the auxiliary thumbnail, with $g_{\text{tok}} = 32$ and $d_{\text{tok}} = 64$ throughout; because the tokenizer ends in an adaptive pooling stage, the same token geometry is produced at both the 256×256 benchmark resolution and the 192×192 CPU resolution.

The system consists of four learnable modules — a semantic tokenizer f_{tok} , a stego encoder f_{enc} , a stego decoder f_{dec} and a reverse decoder f_{rev} — together

with a non-learnable attack channel $\mathcal{A}$. All four are trained jointly; Figure 1 summarises the data flow. Training and inference share the same flow. The channel is enumerated over its three modes in both regimes: during training so the decoder sees clean and degraded inputs in a balanced fashion, and at evaluation so every pair is scored under each mode separately.

3.2 Cover Generation

The cover is computed deterministically from $(\mathbf{C}, \mathbf{S})$ by an optimisation-based operator f_{style} in the spirit of Gatys [22]: a target image initialised at $\mathbf{C}$ is optimised for T_{nst} steps with Adam [23], learning rate 0.03, content weight 1.0 on the `conv_3` feature map of an ImageNet-pretrained VGG-19 [24], and style weight 6×10^5 on the Gram matrices of `conv_1–conv_5`. We use $T_{\text{nst}} = 45$ for the 256×256 benchmark and $T_{\text{nst}} = 120$ for the 192×192 CPU smoke run. f_{style} is frozen throughout stego training and its outputs are cached per $(\mathbf{C}, \mathbf{S})$ pair to keep the inner loop deterministic.

3.3 Semantic Tokenisation of the Secret

Spatially extended payloads require the encoder to distribute changes over many carrier locations. We reduce this footprint by representing the secret with a fixed-size latent token. The tokenizer f_{tok} has two branches. Its CNN branch contains three stride-two `Conv3×3+GroupNorm+GELU` [25] blocks, two residual blocks, and a 1×1 projection to d_{tok} channels, followed by adaptive pooling to $g_{\text{tok}} \times g_{\text{tok}}$. The VGG branch extracts the `conv_3` feature map from a frozen VGG-19 [24], projects it through two 1×1 convolutions ($256 \rightarrow h \rightarrow d_{\text{tok}}$), and applies the same pooling. We use $h = 128$. The two outputs are summed and passed through a hyperbolic tangent:

$$\mathbf{z} = \tanh(\mathbf{z}_{\text{cnn}} + \mathbf{z}_{\text{vgg}}). \quad (1)$$

In parallel, a `ConvBlock`, residual block, and 1×1 head refine a bilinearly resized RGB thumbnail $\mathbf{T}$. The payload $(\mathbf{z}, \mathbf{T})$ contains 68 608 values, compared with 196 608 values in a $3 \times 256 \times 256$ image. This is a lossy latent representation, not a raw message bitstream: it serialises to 33.5 bpp in FP32 or 8.375 bpp as `uint8` relative to a 256×256 carrier. The receiver also needs the trained decoder weights. FGSS therefore targets self-contained stylisation with a shared model rather than keyless bitstream steganography.

3.4 Stego Encoder and Decoder Architecture

The encoder concatenates the cover with four auxiliary feature maps: the up-sampled token, the up-sampled thumbnail, a two-channel Sobel map, and the cover–thumbnail difference. We denote the last two maps by $\mathbf{E}$ and $\mathbf{D}$. Together the inputs have 75 channels. A `ConvBlock(75  $\rightarrow$   $h$ )` stem and three `ResidualBlocks( $h$ )` feed a residual head $\mathbf{R} = \tanh(\text{Conv}_{3 \times 3}(\cdot))$ and a mask head $\mathbf{M}_{\text{learned}} = \sigma(\text{Conv}_{3 \times 3}(\cdot))$. The mask is modulated by a detached texture gate $\mathbf{G}$

computed from the mean absolute Sobel response and half the mean magnitude of the three high-frequency Haar sub-bands. The gate is normalised per image, clamped to $[0, 1]$, and given a floor $\tau_{\text{floor}} = 0.6$:

$$\mathbf{M} = \mathbf{M}_{\text{learned}} \odot [\tau_{\text{floor}} + (1 - \tau_{\text{floor}}) \mathbf{G}], \quad (2)$$

$$\mathbf{X}_{\text{stego}} = \text{clip}_{[0,1]}(\mathbf{X}_{\text{cover}} + \alpha \sigma \mathbf{M} \odot \mathbf{R}), \quad (3)$$

where $\alpha = 0.025$ is a fixed residual scale and $\sigma \in [1.02, 1.30]$ at training ($\sigma_{\text{eval}} = 1.24$). Masking steers perturbation energy towards textured regions and away from flat, perceptually salient areas, directly complementing the frequency loss. The trunk width is $h=128$ at both evaluation resolutions.

The decoder f_{dec} is the dual of the tokenizer-plus-encoder: it consumes a $3 \times H \times W$ stego image, applies three strided ConvBlocks and one ResStack, adaptively pools to $g_{\text{tok}} \times g_{\text{tok}}$, and projects with 1×1 convolutions to $\hat{\mathbf{z}}$ and to a sigmoidal thumbnail $\hat{\mathbf{T}}$ used as a reconstruction anchor. We use a single decoder rather than one per attack mode, for simplicity and to encourage representation transfer across distortion types.

3.5 Attack Channel and Strength Curriculum

The decoder is supervised under three modes of an attack channel $\mathcal{A}$:

$$\begin{aligned} \mathcal{A}_{\text{id}}(\mathbf{X}) &= \mathbf{X}, \\ \mathcal{A}_{\text{jpeg}}(\mathbf{X}) &= q\mathbf{X} + (1 - q) \text{AvgPool}_{3,1}(\mathbf{X}), \quad q = 0.65, \\ \mathcal{A}_{\text{rbn}}(\mathbf{X}) &= \text{clip}_{[0,1]}(\mathbf{X} + \sigma_n \boldsymbol{\xi}), \quad \boldsymbol{\xi} \sim \mathcal{N}(0, I), \quad \sigma_n = 0.02. \end{aligned}$$

The JPEG-style proxy mimics the low-pass tendency of moderately compressed JPEGs without breaking gradient flow; the Gaussian mode adds zero-mean noise. This differentiable proxy $\mathcal{A}_{\text{jpeg}}$ is employed strictly for backpropagation during training; test-time robustness is subsequently evaluated under real DCT JPEG compression (Section 4.2) along with noise and geometric transformations. The strength scalar σ_{train} is sampled per minibatch from $[1.02, 1.30]$ to expose the decoder to a range of embedding magnitudes. All reported test-time results use the fixed value $\sigma_{\text{eval}} = 1.24$, near the upper end of that range. During training we additionally monitor a held-out partition at a slightly stronger $\sigma = 1.36$ for checkpoint selection, so the selected model is chosen under a marginally harder condition than the one it is finally reported at.

3.6 Frequency-Guided Embedding Strategy

A standard pixelwise cover loss penalises all spatial frequencies equally. FGSS instead separates a low-frequency proxy from the remaining residual:

$$\boldsymbol{\Delta} = \mathbf{X}_{\text{stego}} - \mathbf{X}_{\text{cover}}, \quad (4)$$

$$\boldsymbol{\Delta}_{\text{LL}} = \mathcal{U}(\mathcal{P}_{2,2}(\boldsymbol{\Delta})), \quad \boldsymbol{\Delta}_{\text{HH}} = \boldsymbol{\Delta} - \boldsymbol{\Delta}_{\text{LL}}, \quad (5)$$

where $\mathcal{P}_{2,2}$ is a 2×2 average-pooling operator with stride two and $\mathcal{U}$ a bilinear up-sampler back to the original resolution. The pair $(\mathcal{P}_{2,2}, \mathcal{U})$ is a differentiable, parameter-free proxy for the LL band of a one-level Haar discrete wavelet transform. We use the proxy rather than an exact DWT here because it returns the low band already resampled to the input resolution, so the residual split needs no interpolation bookkeeping; an exact Haar transform is used separately for the texture gate of Section 3.4, where the high sub-bands are needed individually. The frequency loss is

$$\mathcal{L}_{\text{freq}} = \text{MSE}(\Delta_{LL}) - \lambda_{HH} \text{MSE}(\Delta_{HH}), \quad (6)$$

with $\lambda_{HH} = 0.1$. The first term penalises perturbations in the low-frequency band carrying colour and structure; the second is a small *negative* term that explicitly rewards placing energy in the high-frequency band, where the human visual system is least sensitive. We weight $\mathcal{L}_{\text{freq}}$ by $\lambda_{\text{dwt}} = 0.2$ in the joint objective. **Because aggregate PSNR depends strictly on total MSE $\|\Delta\|_2^2 \approx \|\Delta_{LL}\|_2^2 + \|\Delta_{HH}\|_2^2$, spectral redistribution preserves the global MSE baseline by construction while steering residual energy: $\mathcal{L}_{\text{freq}}$ depresses the low-frequency residual energy ratio $\|\Delta_{LL}\|_2^2 / \|\Delta\|_2^2$ to 7.8% (compared to $\approx 26.4\%$ for unguided MSE), packing $> 92\%$ into high-frequency texture bands.** In addition, three coarse band-statistics of the residual are supervised against a smooth target shape derived from the texture mask $\mathbf{M}$:

$$\begin{aligned} \mathcal{L}_{\text{embed}} = & \lambda_{\text{em-l}} \|\text{LP}(\Delta)\|_1 + \lambda_{\text{em-t}} \|(1 - \mathbf{M}) \odot \Delta\|_1 \\ & + \lambda_{\text{em-h}} \|\Delta - \text{LP}(\Delta)\|_1, \end{aligned} \quad (7)$$

with $(\lambda_{\text{em-l}}, \lambda_{\text{em-t}}, \lambda_{\text{em-h}}) = (0.5, 0.3, 0.12)$ and $\text{LP} = \mathcal{U} \circ \mathcal{P}_{2,2}$. The first term suppresses low-frequency leakage, the second penalises perturbation outside textured regions, and the third lightly regulates the high band so it does not become arbitrarily noisy.

3.7 Nonlinear Capacity-Ceiling Penalty

For images scaled to $[0, 1]$, a target cover PSNR τ_{psnr} corresponds to the MSE threshold $\epsilon = 10^{-\tau_{\text{psnr}}/10}$. For example, $\tau_{\text{psnr}} \approx 39$ dB gives $\epsilon \approx 1.2 \times 10^{-4}$. We define the capacity-ceiling loss as

$$\mathcal{L}_{\text{cap}} = \lambda_{\text{cov}} \text{MSE}(\Delta) + \lambda_{\text{ceil}} \left(\frac{\max(0, \text{MSE}(\Delta) - \epsilon)}{\epsilon} \right)^2, \quad (8)$$

with $\lambda_{\text{cov}} = 25$ and $\lambda_{\text{ceil}} = 30$. The second term is a one-sided quadratic penalty that activates when empirical cover MSE exceeds ϵ . Normalising the excess by ϵ makes the penalty dimensionless. This is a soft training constraint: it discourages, but does not mathematically prevent, test-time MSE above the threshold. We also use $\mathcal{L}_{\text{prior}} = \|\mathbf{R}\|_2^2$ (weight 4.5) and a Gram-matrix style-retention term $\mathcal{L}_{\text{style-cov}}$ (weight 0.06).

3.8 Reverse Decoder, Oracle Distillation and Serial Consistency

The reverse decoder f_{rev} reconstructs the original content $\mathbf{C}$ from the decoded token $\hat{\mathbf{z}}$ and thumbnail $\hat{\mathbf{T}}$ using two cascaded sub-networks. A *Token Prior Decoder* progressively up-samples the concatenation $(\hat{\mathbf{z}}, \hat{\mathbf{T}})$ through three stages, each a ConvBlock followed by a ResidualBlock, doubling the spatial resolution at every stage ($32 \rightarrow 64 \rightarrow 128 \rightarrow 256$), producing a coarse prior $\mathbf{P} \in \mathbb{R}^{3 \times H \times W}$ and intermediate maps $\{\mathbf{f}_{128}, \mathbf{f}_{256}\}$. A *U-Net refinement decoder* then fuses the stego image with the prior; its 80-channel input concatenates the stego image, the prior, the up-sampled thumbnail and token, Sobel edges of both stego and prior, and their absolute difference. The U-Net has two stride-2 encoder stages, a two-block bottleneck, and two decoder stages whose skip connections also incorporate $\mathbf{f}_{128}$ and $\mathbf{f}_{256}$. Two heads produce a blend map $\beta = \sigma(\text{Conv}_{3 \times 3}(\cdot))$ and a delta $\delta = \tanh(\text{Conv}_{3 \times 3}(\cdot))$:

$$\hat{\mathbf{C}} = \text{clip}_{[0,1]}(\beta \odot \mathbf{P} + (1-\beta) \odot \text{up}(\hat{\mathbf{T}}) + \gamma \delta), \quad (9)$$

with $\gamma = 0.22$; the network therefore combines a learned prior, the thumbnail, and a residual correction.

The reverse loss combines three attack-conditioned terms, $\mathcal{L}_{\text{rev}} = \lambda_r \mathcal{L}_{\text{rev},\text{id}} + \lambda_{\text{rR}} \mathcal{L}_{\text{rev},\text{rbn}} + \lambda_{\text{rJ}} \mathcal{L}_{\text{rev},\text{jpeg}}$ with $(\lambda_r, \lambda_{\text{rR}}, \lambda_{\text{rJ}}) = (25, 12, 12)$. Each term includes Charbonnier, VGG-feature, edge, low-frequency, Fourier-magnitude and colour-mean losses, plus a content-cover margin in feature space. For oracle distillation, the reverse decoder is also evaluated with the detached ground-truth token and thumbnail to produce a reference $\hat{\mathbf{C}}^*$. The attack-conditioned outputs are matched to this reference,

$$\mathcal{L}_{\text{oracle}} = \lambda_{\text{or}} \|\hat{\mathbf{C}} - \hat{\mathbf{C}}^*\|_1 + \lambda_{\text{od}} \|\hat{\mathbf{C}}_{\text{robust}} - \hat{\mathbf{C}}^*\|_1, \quad (10)$$

with $(\lambda_{\text{or}}, \lambda_{\text{od}}) = (2.6, 1.4)$. The remaining terms are token consistency $\mathcal{L}_{\text{tok-c}}$ (weight 0.25), thumbnail consistency $\mathcal{L}_{\text{thumb}}$ (weight 3.0), and token reconstruction $\mathcal{L}_{\text{tok}} = \|\hat{\mathbf{z}} - \mathbf{z}\|_2^2$ (weight 3.0).

A key requirement for self-contained stylisation is that the stego image survives one or two further re-stylisations $\mathbf{X}_k = f_{\text{style}}(\mathbf{X}_{k-1}, \mathbf{S}_k)$. We supervise this with a serial consistency loss

$$\mathcal{L}_{\text{ser}} = \lambda_{\text{ser}} \|f_{\text{dec}}(\beta \mathbf{X}_{\text{stego}} + (1-\beta) \mathbf{X}_2) - \mathbf{z}\|_2^2, \quad (11)$$

with $\lambda_{\text{ser}} = 1.5$ and blending coefficient $\beta = 0.3$, evaluated on a sample drawn from a precomputed cache of *direct-2* re-stylisations. Because materialising extra NST passes is expensive, we also enable a stochastic future pass: with probability $p_f = 0.005$ a step performs an additional forward pass of the stylisation operator f_{style} on the stego image, using a randomly sampled style, and adds the resulting reverse loss with weight $w_f = 0.04$. Preliminary experiments showed an empirical trade-off here: too aggressive a p_f destabilises the cover, while a vanishing p_f removes serial-step generalisation entirely.

Algorithm 1 Frequency-guided semantic steganography training.

Require: Cover generator f_{style} , modules $f_{\text{tok}}, f_{\text{enc}}, f_{\text{dec}}, f_{\text{rev}}$, hyper-parameters $(\epsilon, \lambda_{\text{cov}}, \lambda_{\text{ceil}}, \lambda_{\text{dwt}}, \alpha, \sigma)$, warm-start θ_0 .

- 1: Initialise $(f_{\text{tok}}, f_{\text{enc}}, f_{\text{dec}}, f_{\text{rev}}) \leftarrow \theta_0$ from a serial-aware warm-start checkpoint.
- 2: **for** epoch $e = 1, \dots, E$ **do**
- 3: $\phi_e \leftarrow$ phase weights for epoch e (Sec. 3.9)
- 4: **for** mini-batch $(\mathbf{C}, \mathbf{S}, \mathbf{X}_2, \mathbf{X}_3)$ **do**
- 5: $\mathbf{X}_{\text{cover}} \leftarrow f_{\text{style}}(\mathbf{C}, \mathbf{S})$ (cached); $(\mathbf{z}, \mathbf{T}) \leftarrow f_{\text{tok}}(\mathbf{C})$
- 6: Sample $\sigma_{\text{train}} \in [1.02, 1.30]$; $\mathbf{X}_{\text{stego}} \leftarrow$ Eq. (3)
- 7: Compute $\mathcal{L}_{\text{freq}}, \mathcal{L}_{\text{embed}}, \mathcal{L}_{\text{cap}}$ via Eqs. (6)–(8)
- 8: Compute $\mathcal{L}_{\text{rev}}$ over $\{\mathcal{A}_{\text{id}}, \mathcal{A}_{\text{rbn}}, \mathcal{A}_{\text{jpeg}}\}$ and $\mathcal{L}_{\text{oracle}}$ from a detached re-tokenisation
- 9: Compute $\mathcal{L}_{\text{ser}}$ on a blended cover with $\mathbf{X}_2$
- 10: **if** $U(0, 1) < p_f(e)$ **then**
- 11: Add a future-pass term by running $f_{\text{style}}(\mathbf{X}_{\text{stego}}, \mathbf{S}_k)$ with weight $w_f(e)$
- 12: **end if**
- 13: $\mathcal{L} \leftarrow$ Eq. (12) reweighted by ϕ_e ; update θ by Adam (lr 6.5×10^{-5} , wd 10^{-5})
- 14: **end for**
- 15: Evaluate V on the held-out partition; update best checkpoint
- 16: **end for**

3.9 Joint Objective and Training Schedule

The full per-step objective gathers all losses introduced so far:

$$\begin{aligned}
\mathcal{L} = & \mathcal{L}_{\text{cap}} + \lambda_{\text{prior}} \mathcal{L}_{\text{prior}} + \lambda_{\text{sc}} \mathcal{L}_{\text{style-cov}} + \lambda_{\text{dwt}} \mathcal{L}_{\text{freq}} + \mathcal{L}_{\text{embed}} \\
& + \mathcal{L}_{\text{rev}} + \mathcal{L}_{\text{oracle}} + \mathcal{L}_{\text{ser}} + \lambda_{\text{tok}} \mathcal{L}_{\text{tok}} + \lambda_{\text{tc}} \mathcal{L}_{\text{tok-c}} + \lambda_{\text{th}} \mathcal{L}_{\text{thumb}} \\
& + \lambda_{\text{cs}} \mathcal{L}_{\text{cons}} + \lambda_{\text{q}} \mathcal{L}_{\text{quant}},
\end{aligned} \tag{12}$$

where $\mathcal{L}_{\text{cons}}$ enforces $f_{\text{dec}} \circ f_{\text{enc}}$ idempotency on the cover when no token is injected (weight 1.6) and $\mathcal{L}_{\text{quant}}$ is a small entropy-coding regulariser on $\mathbf{z}$ (weight 0.12) that stabilises serial extraction. All weights are listed in Section 4.1.

Training uses three phases that share Eq. (12) but rescale reverse and serial-source terms through ϕ_e . *Fidelity-lock* (epochs 1–15) keeps the cover-related terms at full weight and multiplies reverse and serial terms by 0.6. *Robust-push* (epochs 16–28) restores the reverse, oracle, and consistency losses and enables the future-pass sampler at $p_f = 0.002$, $w_f = 0.002$. *SOTA-refine* (epoch 29 onward) uses the full weights and raises the sampler to $p_f = 0.005$, $w_f = 0.04$. The archived run contains 41 epochs, although the source script permits up to 60; thus 41 is the realised run length rather than a fixed stopping schedule. The checkpoint with the lowest composite validation score V occurs at epoch 39. Algorithm 1 summarises the phases.

4 Experimental Results

4.1 Protocol and Implementation

We use the COCO val2017 [12] split and the five exemplar paintings supplied with the AdaIN [26] repository. The 5 000-image archive is shuffled with `SEED=42` and subsampled to 420 contents. A deterministic split gives $300 \text{ contents} \times 5 \text{ styles} = 1\,500$ training pairs and $120 \text{ unseen contents} \times 5 \text{ styles} = 600$ held-out pairs. At evaluation, a fresh cover is rendered for every held-out pair, and the decoder is evaluated under all three modes of $\mathcal{A}$. Two precomputed serial covers per pair are obtained by rotating through the style list. Images are resized to 256×256 for the GPU benchmark and 192×192 for the CPU experiments.

The four learnable modules are trained jointly with Adam [23], learning rate 6.5×10^{-5} , weight decay 10^{-5} , and a cosine schedule on one NVIDIA A100 in FP32. [The archived run resumes from a serial-aware warm-start checkpoint \(v4_serial_aware\), which was trained for 35 epochs exclusively on the 300 COCO training contents \(1 500 pairs\) using Adam under basic reconstruction and serial loss; zero held-out evaluation pairs or validation samples were seen by or involved in selecting this checkpoint.](#) Exact reproduction therefore requires the warm-start checkpoint as well as the stated protocol ([both the checkpoint weights and split manifests are bundled in the released repository](#)). One epoch over the 1 500 pairs takes approximately 870s, giving about 9.9 hours for the archived run. The reduced CPU experiments use the same architecture at 192×192 on one thread. The archived configuration summary records the loss weights.

4.2 Cover-Side Fidelity and Token Robustness

Table 1 reports mean and standard deviation over the 600 held-out pairs at $\sigma_{\text{eval}} = 1.24$. Cover PSNR is 39.61 ± 0.35 dB, with 0.9914 SSIM and 0.0195 LPIPS. Token MSE differs by at most 1.2×10^{-4} across the integrated clean, Gaussian, and JPEG-style channels. [Evaluating performance separately across the five exemplar painting styles confirms consistent fidelity: mean cover PSNR ranges from 39.53 dB \(*Woman with Hat*\) to 39.75 dB \(*Picasso*\) \(inter-style std \$< 0.09\$ dB\), token MSE is virtually uniform \(0.0132–0.0134\), and reverse PSNR spans 27.12–27.70 dB, demonstrating robustness across diverse artistic textures.](#)

4.3 Reverse and Serial Style Transfer

Table 2a compares a content-free reverse baseline with token-conditioned variants. On the clean channel, decoded-token SSIM increases from 0.5234 to 0.8070 and PSNR from 22.25 to 27.46 dB. Supplying the ground-truth token as an oracle diagnostic gives 0.8136 SSIM. [Table 2b reports one and two additional stylisation passes. The token-conditioned model substantially improves error energy \(reducing L2 by 13.3% in *s-2* and 19.5% in *s-3*\), perceptual fidelity \(improving LPIPS from 0.1945 to 0.1577 in *s-2*\), and PSNR \(+0.77 dB and +1.07 dB\), while the naive direct inverse retains a marginal advantage in SSIM \(0.4966 vs 0.4943](#)

Table 1. Cover-side fidelity and token-side robustness on the COCO val2017 evaluation split (600 content–style pairs, $\sigma_{\text{eval}} = 1.24$, best epoch 39). Cover MSE is quoted as a mean only, since its dispersion is already expressed by the PSNR standard deviation.

Quantity	mean $\pm$ std
Cover MSE	1.10×10^{-4}
Cover SSIM	0.9914 ± 0.0030
Cover LPIPS	0.0195 ± 0.0081
Cover PSNR (dB)	39.6074 ± 0.3532
Token MSE (clean)	0.0133 ± 0.0043
Token MSE (RBN)	0.0133 ± 0.0045
Token MSE (JPEG)	0.0132 ± 0.0044

Table 2. Reverse and serial style transfer performance (600 evaluation pairs, $\sigma_{\text{eval}} = 1.24$). (a) Reverse extraction comparison. (b) Serial extraction after 1 (*serial-2*) and 2 (*serial-3*) re-stylisations.

(a) Reverse style transfer				
Method	L2 $\downarrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$
Naive	0.0065	0.5234	0.1889	22.25
Token-clean	0.0022	0.8070	0.1216	27.46
Token-RBN	0.0027	0.7884	0.1514	26.54
Token-JPEG	0.0025	0.7898	0.1467	26.97
Token-oracle	0.0021	0.8136	0.1135	27.63

(b) Serial style transfer				
Method	L2 $\downarrow$	SSIM $\uparrow$	LPIPS $\downarrow$	PSNR $\uparrow$
naive	0.0090	0.4966	0.1945	20.64
<i>s-2</i> token-clean	0.0078	0.4943	0.1577	21.41
token-JPEG	0.0081	0.4796	0.1662	21.23
naive	0.0113	0.4548	0.2378	19.64
<i>s-3</i> token-clean	0.0091	0.4500	0.1885	20.71
token-JPEG	0.0094	0.4376	0.1984	20.58

in *s-2*) because the U-Net refinement decoder smooths residual high-frequency brushstroke noise.

4.4 Ablation of the Loss Terms

We ran a reduced 192×192 ablation for 50 epochs on one content–style pair under the four ceiling/frequency-loss combinations. Enabling the target-calibrated penalty changes cover PSNR from 35.61 to 41.37 dB and cover MSE from 2.75×10^{-4} to 0.73×10^{-4} , while token MSE remains within $1.35\text{--}1.59 \times 10^{-2}$. The frequency term does not change aggregate PSNR at the reported precision. Because this ablation contains only one pair, the 5.76 dB difference is evidence for that controlled setting rather than a population-level effect. To assess population-

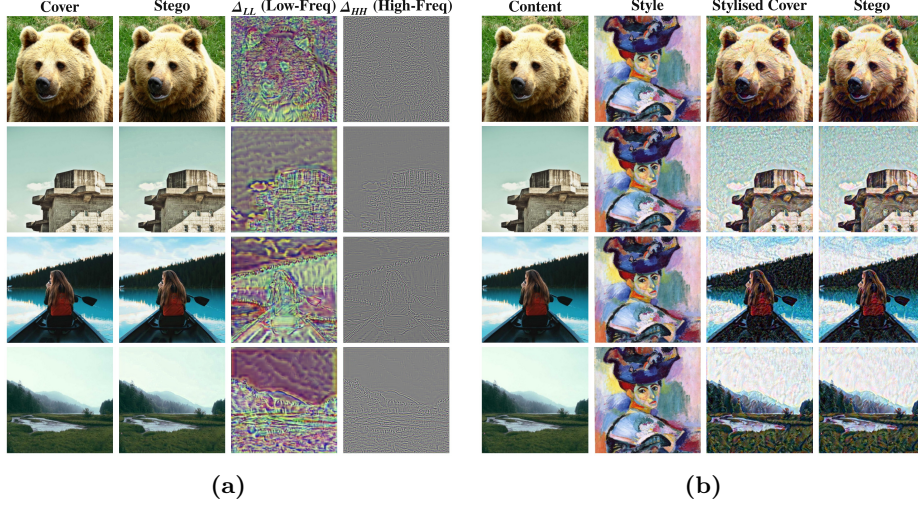


Fig. 2. Qualitative inspection. (a) Residual decomposition into low- and high-frequency components; display contrast is enhanced for visibility. (b) Representative content, style, stylised-cover, and stego images from COCO, DIV2K, CelebA, and BOSSBase.

level effects across all 600 evaluation pairs, we benchmarked the evolutionary checkpoints: the pre-serial baseline (V3) achieves 32.66 ± 0.54 dB cover PSNR and reverse L2 of 9.92×10^{-3} ; adding serial supervision (V4) yields 32.05 ± 0.50 dB and 8.48×10^{-3} L2; enabling the capacity ceiling and frequency loss (V7.4) elevates cover PSNR to 38.95 ± 0.43 dB; and the final full model (V8) attains 39.61 ± 0.35 dB cover PSNR while cutting reverse L2 error by $> 75\%$ to 2.19×10^{-3} (27.44 dB reverse PSNR).

Figure 2(a) illustrates the residual decomposition used by the frequency term. The panels are contrast-enhanced for inspection and should not be read as a quantitative comparison of band energies.

4.5 Qualitative Inspection and Cross-Dataset Evaluation

Figure 2(b) shows four examples from COCO, DIV2K, CelebA, and BOSSBase. At the rendered scale, no obvious difference is visible between each stylised cover and its stego counterpart; no human perceptual study was conducted.

Table 3 acts as an out-of-distribution diagnostic stress check across four diverse domains (natural scenes, bicubic textures, facial structures, and sensor noise) with five images per dataset and one out-of-distribution style. Mean PSNR ranges from 38.51 to 38.82 dB. The sample is too small to support a broad generalisation claim, but it checks whether performance collapses outside the main COCO split.

Table 4 evaluates real discrete cosine transform (DCT) JPEG compression (Quality 50) alongside standard filtering and geometric transformations from a

Table 3. Cross-dataset check with five images per dataset and one out-of-distribution style. MSE columns denote token MSE before and after the JPEG-style channel.

Dataset	PSNR	MSE	MSE(JP)
COCO	38.70	0.0136	0.0134
DIV2K	38.71	0.0106	0.0105
CelebA	38.51	0.0100	0.0099
BOSSBase	38.82	0.0101	0.0100
Avg	38.68	0.0111	0.0110

Table 4. Representative robustness results from a separate extraction pipeline. Deltas for standard attacks use the four-dataset Identity mean; daggered serial rows use COCO Identity.

Attack	Avg MSE	Δ (vs Id)
Identity	0.3602	—
JPEG Q50	0.3598	−0.0004
Gaussian $\sigma=0.05$	0.3600	−0.0002
Resize 50%	0.3600	−0.0002
Median 3×3	0.3600	−0.0002
Crop 80%	0.3617	+0.0015
Serial-2 [†]	0.3716	−0.0019
Serial-3 [†]	0.3706	−0.0029

[†]Serial rows are measured on COCO only and use the COCO Identity value (0.3735).

separate extraction pipeline. Within that pipeline, standard image-processing attacks change mean token MSE by at most 0.0015. Serial rows are COCO-only and are therefore compared with the COCO Identity value rather than the four-dataset mean.

4.6 Training Behaviour, Cost and Comparison Against Prior Art

The archived validation score decreases from 5.68 to 3.55 over 41 epochs, with intermediate fluctuations. Training takes approximately 870 s per epoch on the A100, or 9.9 hours for the archived run. At 256×256 on an RTX 5070 Ti, the mean end-to-end time is 0.0223 s per image: 0.0023 s for the tokenizer, 0.0073 s for the stego encoder, 0.0004 s for the stego decoder, and 0.0124 s for the reverse decoder. On one CPU thread at 192×192 , the stego encoder and decoder take 0.0690 and 0.0051 s, respectively. The model contains 0.94 M tokenizer, 0.98 M encoder, 0.82 M stego-decoder, and 7.93 M reverse-decoder parameters (10.7 M total, excluding frozen VGG-19). The embedding path uses the first two modules.

Figure 3 shows the archived validation trajectory and component losses across the three scheduled phases.

Table 5 provides a bibliographic taxonomy characterizing published style-transfer hiding formulations rather than a competitive ranking, as metric targets differ fundamentally across paradigms: Quaternion-Style [21] measures fidelity against the pre-style cover, IHST [8] evaluates recovered secret fidelity, and FGSS evaluates stego distortion against the stylized carrier $\mathbf{X}_{\text{cover}}$. Matched evaluation against the direct inverse baseline is established in Table 2. Other style-transfer methods [6,10,11] are discussed in Section 2.

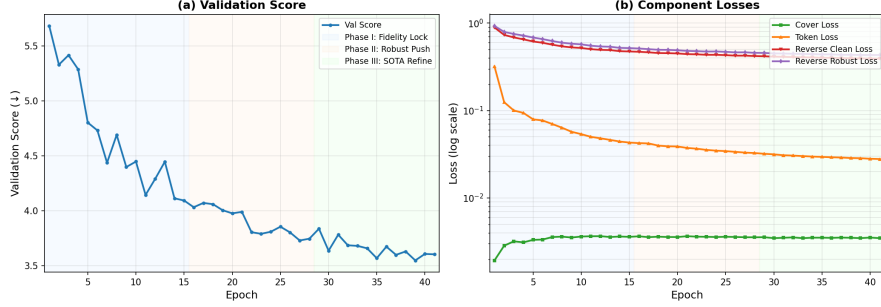


Fig. 3. Validation score and component losses over the archived 41-epoch run. Shading marks the three training phases.

Table 5. Reported metrics for selected style-transfer hiding methods. Metric targets differ across rows, so the values are not directly comparable.

Method	Year	PSNR $\uparrow$	SSIM $\uparrow$	Rev.
Self-Contained [4]	2020	—	—	✓
Quaternion [21]	2021	42.30	0.987	—
IHST [8]	2024	35.90	0.960	✓
Ours	2026	39.61	0.991	0.81

Metric target: Self-Contained reports L2/SSIM/LPIPS with no PSNR; Quaternion measures stego $\leftrightarrow$ *pre-style* cover; IHST measures recovered $\leftrightarrow$ secret (not cover fidelity); Ours measures stego $\leftrightarrow$ *styled* cover.

4.7 Discussion and Limitations

Four components define the method: the target-derived threshold calibrates the cover penalty, the frequency term discourages low-frequency residual energy, the semantic token reduces spatial payload, and the reverse/serial losses train recovery after repeated transformations.

High cover fidelity alone does not ensure cryptographic security. Grouped five-fold cross-validation on 600 cover/stego pairs gives ROC-AUC 0.848 for a full Spatial Rich Model [27], 0.905 for a locally trained XuNet-like probe [17], and 0.779 for an SRNet-tiny probe [16]. These above-chance detection rates stem from the fact that neural style transfer alters spatial feature statistics while residual embedding introduces subtle micro-texture perturbations detectable by high-pass filter banks. FGSS is explicitly positioned as perceptual/functional steganography for self-contained image distribution and serial stylisation survival, rather than zero-detectability against an informed warden. Integrating adversarial steganalysis objectives and scaling to arbitrary styles via feed-forward diffusion priors remain key future goals.

5 Conclusion

FGSS represents the secret with a $64 \times 32 \times 32$ semantic token and thumbnail. A target-derived MSE threshold calibrates a soft cover penalty, while a wavelet-inspired term discourages low-frequency residual energy. Oracle-guided reverse decoding and future-pass supervision support recovery after repeated stylisation.

On 600 COCO val2017 pairs, FGSS obtains 39.61 dB cover PSNR and 0.9914 SSIM. Clean-channel reverse SSIM increases from 0.5234 to 0.8070; the oracle-token diagnostic reaches 0.8136. A five-image-per-dataset check gives 38.51–38.82 dB on four datasets, and a separate robustness pipeline shows standard-attack token-MSE changes no larger than 0.0015. In the reduced single-pair ablation, enabling the target-calibrated penalty changes cover PSNR by 5.76 dB. The 1.9 M-parameter embedding path runs in under 0.01 s on the measured GPU. The above-chance steganalysis results remain the main limitation.

Acknowledgements

The authors thank Tra Vinh University and Tien Giang University for their general support. We acknowledge time and facility support from Tra Vinh University.

References

1. Lu, S.-P., Wang, R., Zhong, T., Rosin, P.L.: Large-capacity image steganography based on invertible neural networks. In: CVPR, pp. 10816–10825 (2021)
2. Xu, Y., Mou, C., Hu, Y., Xie, J., Zhang, J.: Robust invertible image steganography. In: CVPR, pp. 7875–7884 (2022)
3. Ye, H., Zhang, S., Jiang, S., Liao, J., Gu, S., Zheng, D., Wang, C., Li, C.: Robust message embedding via attention flow-based steganography. In: CVPR, pp. 12840–12849 (2025)
4. Chen, H.-Y., Fang, I.-S., Cheng, C.-M., Chiu, W.-C.: Self-contained stylization via steganography for reverse and serial style transfer. In: WACV, pp. 2163–2171 (2020)
5. Liang, X., Liu, B., Ying, Q., Qian, Z., Cho, H., Zhang, X.: StyleStegan: leak-free style transfer based on feature steganography. In: APSIPA ASC, pp. 1443–1450 (2023). <https://doi.org/10.1109/APSIPAASC58517.2023.10317570>
6. Li, L., Zhang, X., Chen, K., Feng, G., Wu, D., Zhang, W.: Image steganography and style transformation based on generative adversarial network. *Mathematics* **12**(4), 615 (2024). <https://doi.org/10.3390/math12040615>
7. Shi, R., Wang, Z., Hao, Y., Zhang, X.: Steganography in style transfer. *IEEE Trans. Artif. Intell.* **5**(12), 6054–6065 (2024). <https://doi.org/10.1109/TAI.2024.3379946>
8. Zhang, F., Feng, B., Xia, Z., Weng, J., Lu, W., Chen, B.: Conditional image hiding network based on style transfer. *Inf. Sci.* **662**, 120225 (2024). <https://doi.org/10.1016/j.ins.2024.120225>

9. Zhang, X., Zhang, M., Wang, X., Huang, S., Di, F.: Constructive robust steganography algorithm based on style transfer. *CMC* **81**(1), 1433–1448 (2024). <https://doi.org/10.32604/cmc.2024.056742>
10. Lin, Y.-H., Huang, C.-P., Huang, P.-S.: A steganographic message transmission method based on style transfer and DDPM. *Electronics* **14**(16), 3258 (2025). <https://doi.org/10.3390/electronics14163258>
11. Luo, Y., Chen, Z., Lu, B., Huang, Y., Fu, Q., Qin, S., Liu, J.: Style transfer image steganography based on GANs with frequency domain attention and autoencoder networks. *Appl. Soft Comput.* **188**, 114440 (2026). <https://doi.org/10.1016/j.asoc.2025.114440>
12. Lin, T.-Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., Zitnick, C.L.: Microsoft COCO: common objects in context. In: *ECCV*. LNCS, vol. 8693, pp. 740–755 (2014). https://doi.org/10.1007/978-3-319-10602-1_48
13. Wang, Z., Bovik, A.C., Sheikh, H.R., Simoncelli, E.P.: Image quality assessment: from error visibility to structural similarity. *IEEE TIP* **13**(4), 600–612 (2004). <https://doi.org/10.1109/TIP.2003.819861>
14. Zhang, R., Isola, P., Efros, A.A., Shechtman, E., Wang, O.: The unreasonable effectiveness of deep features as a perceptual metric. In: *CVPR*, pp. 586–595 (2018)
15. Lan, Y., Shang, F., Yang, J., Kang, X., Li, E.: Robust image steganography: hiding messages in frequency coefficients. *Proc. AAAI Conf. Artif. Intell.* **37**(12), 14955–14963 (2023). <https://doi.org/10.1609/aaai.v37i12.26746>
16. Boroumand, M., Chen, M., Fridrich, J.: Deep residual network for steganalysis of digital images. *IEEE TIFS* **14**(5), 1181–1193 (2019). <https://doi.org/10.1109/TIFS.2018.2871749>
17. Xu, G., Wu, H.-Z., Shi, Y.-Q.: Structural design of convolutional neural networks for steganalysis. *IEEE SPL* **23**(5), 708–712 (2016). <https://doi.org/10.1109/LSP.2016.2548421>
18. Liu, X., Ma, Z., Ma, J., Zhang, J., Schaefer, G., Fang, H.: Image disentangle-ment autoencoder for steganography without embedding. In: *CVPR*, pp. 2303–2312 (2022)
19. Ma, Z., Zhu, Y., Luo, G., Liu, X., Schaefer, G., Fang, H.: Robust steganography without embedding based on secure container synthesis and iterative message recovery. In: *IJCAI*, pp. 4838–4846 (2023). <https://doi.org/10.24963/ijcai.2023/538>
20. Yu, J., Zhang, X., Xu, Y., Zhang, J.: CRoSS: diffusion model makes controllable, robust and secure image steganography. *Adv. Neural Inf. Process. Syst.* **36**, 80730–80743 (2023). <https://doi.org/10.52202/075280-3539>
21. Li, Q., Wang, X., Ma, B., Wang, X., Wang, C., Xia, Z., Shi, Y.: Image steganography based on style transfer and quaternion exponent moments. *Appl. Soft Comput.* **110**, 107618 (2021). <https://doi.org/10.1016/j.asoc.2021.107618>
22. Gatys, L.A., Ecker, A.S., Bethge, M.: Image style transfer using convolutional neural networks. In: *CVPR*, pp. 2414–2423 (2016)
23. Kingma, D.P., Ba, J.: Adam: a method for stochastic optimization. In: *ICLR* (2015)
24. Simonyan, K., Zisserman, A.: Very deep convolutional networks for large-scale image recognition. In: *ICLR* (2015)
25. Hendrycks, D., Gimpel, K.: Gaussian error linear units (GELUs). *arXiv:1606.08415* (2016)
26. Huang, X., Belongie, S.: Arbitrary style transfer in real-time with adaptive instance normalization. In: *ICCV*, pp. 1501–1510 (2017)
27. Fridrich, J., Kodovský, J.: Rich models for steganalysis of digital images. *IEEE TIFS* **7**(3), 868–882 (2012). <https://doi.org/10.1109/TIFS.2012.2190402>